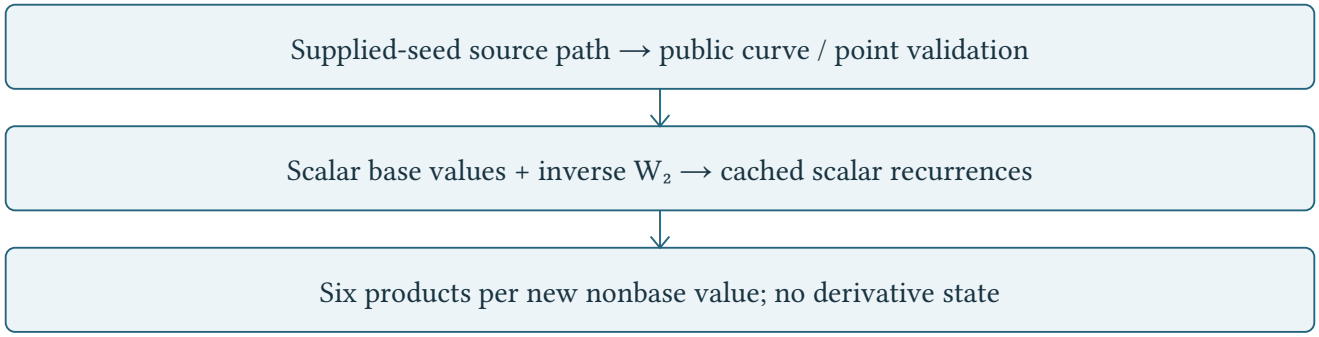
Value-only native division-polynomial evaluation

2026-11-03 · Exact software checks · Derived recurrence cost

The new ValueDivision core computes scalar division values without derivative arithmetic or storage. It is intended for source calls whose normalization seed is already supplied by the outer recurrence. The prior checked jet core remains available for actual normalization and derivative requests.



Jet core remains for normalization | Full source integration is next

Construction. The standard even/odd recurrences are unchanged. Explicit squares and cubes require exactly six field multiplications and one subtraction per newly generated nonbase node, after dependencies are available. Inverse W_2 is prepared once. Cache hits perform no recurrence arithmetic.

Accounting. The six-product count excludes curve validation, base polynomial construction, inversion, map lookup and storage overhead. Scalar wrappers also retain the modulus. No elapsed-time improvement or complete-source cost is measured. The two engines have separate caches; constructing both duplicates setup and must be charged if a caller chooses that path.

Native checks. At sourceb3b7dd54a375, one rustc build and one selected-test invocation succeeded. Three new methods passed; three historical jet methods were filtered out. All9 saved division values match. On the three original generators and their negatives, scalar and jet results agree at indices0..63, r, (r+1)/2 and u64::MAX; independent antipodal parity also passes. Domain, point-membership and torsion-zero controls pass.

Scope. The public API takes no target scalar or precomputed sign. The production path has no jet dependency; the old module is a test reference. Prime p is caller-supplied and bounded by half the u64 maximum. Maximum-index checks do not establish arbitrary-size field arithmetic. No fresh sign source, recovery cohort, cloud job or benchmark ran.

Next. Route supplied-seed calls to ValueDivision and normalization to jets; port geometric progression/resultant stages and connect the sign-mask decoder. Retain the full original4,563-case cohort and admit its native runtime before execution. These unit checks do not substitute for the cohort.

Evidence. lib.rs, check-lock.sha256, check-receipt.txt and build/test logs. DERIVATION.md details cost exclusions, exact test scope and provenance.

Reference. Mathlib elliptic divisibility sequences, and the preserved native jet core and public setup fixtures. Standard recurrences are specialized; no novelty or statistical claim is made.